Emergent Topological and Optical Properties in Pristine and Hydrogen-Functionalized Two-Dimensional Beryllium Phases

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Abstract

Two-dimensional elemental materials provide a unique platform for exploring electronic, optical, and topological phenomena arising from reduced dimensionality and modified bonding environments. Here, using density functional theory calculations, we investigate the structural stability, electronic structure, optical response, and topological characteristics of pristine and hydrogen-functionalized beryllene systems, including monolayer α-beryllene, bilayer β-beryllene, and a newly identified cubic trilayer phase, together with bulk reference phases. In addition to confirming the stability of previously proposed α- and β-beryllene phases, we predict a dynamically stable cubic trilayer beryllene phase with distinct electronic and optical characteristics. Hydrogen functionalization substantially modifies the structural and electronic properties, yielding stable hydrogen-terminated configurations and driving α-beryllene into a wide-gap semiconducting state, while preserving metallic behaviour in hydrogen-functionalized β-based structures. The calculated dielectric response reveals strong anisotropy and pronounced low-dimensional optical effects, including enhanced visible-to-ultraviolet absorption, modified plasmonic behaviour, and direction-dependent dielectric responses associated with collective electronic excitations. Furthermore, parity analysis of the spin-orbit-coupled electronic states demonstrates that, beyond previously reported α-beryllene, the cubic trilayer phase also exhibits a non-trivial Z_2 topological invariant, indicating the emergence of topologically protected electronic states in a light-element system. These results highlight beryllene as an unusual elemental two-dimensional platform combining metallicity, tunable optical properties, and non-trivial topology, with potential relevance for future nanoscale electronic, photonic, and quantum technologies.

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I. INTRODUCTION

Two-dimensional materials have reshaped modern materials science by enabling access to electronic, thermal, and mechanical properties that differ substantially from those of their bulk counterparts. Since the isolation of graphene, reduced dimensionality has been shown to give rise to Dirac fermions, enhanced carrier mobility, strong many-body interactions, tuneable band structures, and anomalous phonon transport [1, 2]. These effects support rapid progress in nanoelectronics, optoelectronics, sensing, thermal management, and quantum technologies.

Beyond graphene, the discovery and prediction of atomically thin materials has expanded to include transition-metal dichalcogenides, hexagonal boron nitride, black phosphorus, and elemental two-dimensional materials, commonly referred to as Xenes [3, 4]. Elemental two-dimensional materials are particularly attractive because their chemically simple nature allows intrinsic structure-property relationships to be studied without the complexity introduced by multicomponent bonding. However, most experimentally realized Xenes are composed of relatively heavy elements, where strong spin-orbit coupling, substrate effects, or structural buckling dominate the physics and may obscure low-energy electronic behaviour.

In contrast, beryllium represents an extreme and largely unexplored limit within the family of elemental two-dimensional materials. As the lightest alkaline-earth metal, bulk beryllium exhibits a unique combination of low density, high elastic modulus, high thermal conductivity, and transparency to X-rays and neutrons. These properties have led to its use in aerospace, nuclear, and optical applications [5]. Extending elemental beryllium into the two-dimensional limit raises the prospect of ultralight metallic layers with unconventional electronic and thermal properties governed by reduced dimensionality and strong confinement.

First-principles studies predicted that atomically thin beryllium could exist in metastable hexagonal or stable two-layer hexagonal lattice configurations [6, 7]. However, for many years, two-dimensional beryllium remained a theoretical construct due to experimental challenges associated with synthesis and handling. This situation changed recently with the experimental realization of two-dimensional beryllium, termed beryllene. Chahal *et al.* reported the first experimental synthesis and characterization of beryllene using a sonochemical exfoliation approach starting from elemental beryllium powder [8]. High-resolution

transmission electron microscopy revealed atomically thin sheets with multiple structural motifs, including hexagonal and square-like lattice geometries. Raman spectroscopy and electron diffraction confirmed crystallinity, while electrical measurements demonstrated intrinsic metallic conductivity.

Subsequent theoretical investigations have uncovered striking thermal transport behaviour in hexagonal monolayer beryllene. Using first-principles phonon calculations combined with the Boltzmann transport equation, Chowdhury and Mogurampelly predicted an exceptionally high in-plane lattice thermal conductivity of approximately $270\,\mathrm{W\,m^{-1}\,K^{-1}}$ at room temperature, which exceeds that of bulk beryllium [9]. This enhancement was attributed to suppressed phonon-phonon scattering and the dominant contribution of long-wavelength flexural phonons, a feature rarely observed in atomically thin metallic systems. These results suggest that two-dimensional beryllium could serve as an efficient heat-spreading material in nanoscale electronic architectures.

Beyond thermal properties, the electronic structure of beryllene has been predicted to host non-trivial quantum features. Several density functional theory studies have identified Dirac-like band crossings near the Fermi level in hexagonal monolayer and bilayer beryllene, usually denoted as α- and β-beryllene, arising from symmetry-protected s-p orbital hybridization [10]. Electron-phonon coupling calculations further suggest that these phases may support phonon-mediated superconductivity with critical temperatures on the order of $10\,\mathrm{K}$. The coexistence of Dirac fermions and superconductivity in an ultralight elemental monolayer distinguishes beryllene from both graphene and heavier Xenes, and highlights its potential as a platform for low-dimensional quantum phenomena.

The possibility of topological behaviour in two-dimensional beryllium has also been explored from a more fundamental perspective. Derriche *et al.* demonstrated that light-element lattices dominated by charge-based interactions can host topologically protected electronic states even in the absence of strong spin-orbit coupling [11]. These findings suggest that beryllene may realize topological phases driven mainly by lattice symmetry and orbital physics, offering an alternative route to low-dissipation electronic states without relying on heavy elements. While experimental studies of beryllene remain limited, earlier work on ultrathin beryllium films and nanostructures has revealed unconventional quantum transport behaviour, including strong disorder effects and superconducting correlations at low temperatures [12].

Two-dimensional topological materials are considered promising for quantum and electronic technologies because their non-trivial band topology gives rise to robust, symmetry-protected edge states that resist disorder and backscattering [13]. Recent density functional theory studies have shown that simple elemental two-dimensional systems can host switchable or adsorbate-induced topological behaviour. For example, interlayer sliding in bilayer elemental bismuth was predicted to switch the system between topologically trivial and non-trivial phases [14]. Similarly, hydrogenation and strain can tune monolayer arsenene into a quantum spin Hall state with a sizable band gap [15]. These studies indicate that structural control and adsorbate functionalization can provide practical routes to topological phases in elemental two-dimensional materials.

Although two-dimensional elemental beryllium has recently been realized experimentally and shown to exhibit unusual electronic and thermal properties, its interaction with hydrogen has received much less attention. First-principles investigations of hydrogen adsorption on low-index hexagonal close-packed beryllium surfaces, particularly Be(0001), show that hydrogen binds preferentially to high-symmetry adsorption sites with moderate binding energies and induces appreciable local lattice relaxation [16]. These results indicate that beryllium-hydrogen interactions occupy an intermediate regime between weak physisorption and strong ionic hydride formation. In the two-dimensional limit, reduced coordination and enhanced lattice flexibility are expected to modify this balance, potentially stabilizing adsorption geometries or electronic reconstructions that are inaccessible on bulk surfaces. Using global structure search and density functional theory calculations, two stable beryllium hydride monolayers were predicted with multicentre beryllium-hydrogen bonding motifs and wide band gaps [17]. Although such systems correspond to compound formation rather than surface termination, they demonstrate that hydrogen can strongly reshape bonding and electronic behaviour in low-dimensional beryllium-based systems.

In this work, we use density functional theory calculations to investigate the atomic, electronic, optical, and topological properties of pristine and hydrogen-functionalized beryllene systems. The results identify stable hydrogen-functionalized structures and a cubic trilayer geometry with non-trivial Z_2 topology. The optical properties demonstrate direction-dependent and dimensionality-dependent response, together with plasmon-related features and visible-to-ultraviolet absorption.

II. COMPUTATIONAL METHODS

The density functional theory calculations are performed using the Vienna Ab initio Simulation Package, VASP [18–21]. The projector augmented wave method is used to describe the electron-ion interactions [22]. The generalized gradient approximation of Perdew–Burke–Ernzerhof is employed for the exchange-correlation functional [23], along with the D3 dispersion correction with Becke–Johnson damping to account for long-range van der Waals interactions [24–26].

For bulk hexagonal close-packed beryllium and bulk body-centred cubic beryllium, a Γ-centered k-point mesh of $27 \times 27 \times 27$ is used for the total energy and structural relaxation calculations. A $105 \times 105 \times 105$ mesh is used for the calculation of the complex dielectric function. A plane-wave energy cutoff of 600 eV is used for all calculations except the topological calculations described below. The electronic convergence criterion is set to 1×10^{-6} eV, and the structures are relaxed until all forces on each atom are below 0.01 eV Å^{-1}. For the two-dimensional systems, a vacuum region of 40 Å is included. A k-point mesh of $27 \times 27 \times 1$ is used for the hexagonal structures, whereas a $25 \times 25 \times 1$ mesh is used for the cubic structures. For all two-dimensional systems, a $105 \times 105 \times 1$ mesh is used for the calculation of the complex dielectric function. Convergence tests for the cohesive and total energies, as well as for the dielectric function with respect to the k-point set and the number of bands included through `NBANDS`, are provided in figures S1–S12 of the Supplementary Information.

The dynamical stability of the structures is examined using finite-difference phonon calculations with 5×5 supercells. The finite-temperature stability of the hydrogen-functionalized structures is further examined using *ab initio* molecular dynamics simulations.

The adsorption energy per hydrogen atom, E_{ad}, is calculated as:

$$E_{\mathrm{ad}} = \frac{E_{\mathrm{H/Be}} - E_{\mathrm{Be}} - E_{\mathrm{H}}}{n_{\mathrm{H}}} \tag{1}$$

here, $E_{\mathrm{H/Be}}$ is the total energy of the hydrogen-functionalized system, E_{Be} is the total energy of the corresponding pristine two-dimensional beryllium structure, E_{H} is the energy per hydrogen atom in an isolated H_2 molecule, and n_{H} is the number of hydrogen atoms in the unit cell.

The cohesive energy per atom, E_{c}, of the beryllium systems is calculated as:

$$E_{\mathrm{c}} = \frac{E_{\text{Be-struc}} - n_{\mathrm{Be}} E_{\text{Be-atom}}}{n_{\mathrm{Be}}} \tag{2}$$

here, $E_{\text{Be-struc}}$ is the total energy of the beryllium structure of interest, $E_{\text{Be-atom}}$ is the total energy of a free beryllium atom calculated in a large supercell, and n_{Be} is the number of beryllium atoms in the unit cell.

For the topological calculations, we use an energy cutoff of 520 eV, a $24 \times 24 \times 1$ k-point mesh, and a vacuum region of 15 Å. The self-consistent convergence criterion is set to 1×10^{-7} eV, and Gaussian smearing of 0.02 eV is used. Spin-orbit coupling is included in the band-structure and parity calculations. The parity eigenvalues of the spinor wave functions are calculated using `irvsp` [27]. For a time-reversal invariant momentum Λ_i, the parity product is given by:

$$\delta_i = \prod_{m=1}^{N_{\mathrm{occ}}/2} \xi_{2m}(\Lambda_i) \tag{3}$$

here, $\xi_{2m}(\Lambda_i) = \pm 1$ is the parity eigenvalue of one state in the mth Kramers pair at Λ_i. The Z_2 index is then obtained from the Fu–Kane product [28, 29]:

$$(-1)^{\nu} = \prod_i \delta_i \tag{4}$$

here, $\nu = 1$ denotes non-trivial topology, whereas $\nu = 0$ denotes trivial topology.

For the optical calculations, we calculate the frequency-dependent dielectric tensor $\epsilon_{\alpha\beta}(\omega)$ within VASP. The complex dielectric tensor is written as:

$$\epsilon_{\alpha\beta}(\omega) = \epsilon_{1,\alpha\beta}(\omega) + i\epsilon_{2,\alpha\beta}(\omega) \tag{5}$$

here, $\epsilon_{1,\alpha\beta}(\omega)$ and $\epsilon_{2,\alpha\beta}(\omega)$ are the real and imaginary parts, respectively, and ω denotes the photon energy. Within the long-wavelength independent-particle approximation, the imaginary part is calculated as [30, 31]:

$$\begin{aligned}
\epsilon_{2,\alpha\beta}(\omega) &= \operatorname{Im} \epsilon_{\alpha\beta}(\omega) \\
&= \frac{4\pi^2 e^2}{\Omega} \lim_{q\to 0} \frac{1}{q^2} \sum_{c,v,\mathbf{k}} 2\omega_{\mathbf{k}} \delta(E_{c\mathbf{k}} - E_{v\mathbf{k}} - \omega) \\
&\quad \times \left\langle u_{c,\mathbf{k}+\mathbf{e}_\alpha q} | u_{v\mathbf{k}} \right\rangle \left\langle u_{v\mathbf{k}} \middle| u_{c,\mathbf{k}+\mathbf{e}_\beta q} \right\rangle
\end{aligned} \tag{6}$$

here, $\mathbf{k}$ is the electron wavevector, $\omega_{\mathbf{k}}$ is the k-point weight, q is the Bloch vector of the incident wave, and Ω is the unit cell volume. The indices v and c denote the valence and conduction bands, respectively; $u_{v\mathbf{k}}$ and $u_{c\mathbf{k}}$ are the cell-periodic parts of the corresponding orbitals; $\mathbf{e}_\alpha$ and $\mathbf{e}_\beta$ are unit vectors along the Cartesian directions; and $E_{v\mathbf{k}}$ and $E_{c\mathbf{k}}$ are the valence band and conduction band energies.

The real part is obtained by applying the Kramers–Kronig transformation:

$$\epsilon_{1,\alpha\beta}(\omega) = \mathrm{Re}\,\epsilon_{\alpha\beta}(\omega) = \delta_{\alpha\beta} + \frac{2}{\pi}\mathcal{P}\int_0^\infty \mathrm{d}\omega' \frac{\omega'\epsilon_{2,\alpha\beta}(\omega')}{\omega'^2 - \omega^2 + i\eta} \tag{7}$$

here, $\delta_{\alpha\beta}$ is the Kronecker delta, $\mathcal{P}$ denotes the Cauchy principal value, and η is the complex shift, with $\eta = 0.1$ in the present calculations.

Along a principal direction λ, the tensor indices are suppressed and the complex refractive index is written as $N_\lambda(\omega) = n_\lambda(\omega) + ik_\lambda(\omega)$, with $N_\lambda^2(\omega) = \epsilon_{\lambda\lambda}(\omega)$. The absorption coefficient, refractive index, extinction coefficient, reflectivity, and energy-loss spectrum are then obtained from the dielectric function as follows [30]:

$$\alpha_\lambda(\omega) = \frac{\sqrt{2}\omega}{c}\left[\sqrt{\epsilon_{1,\lambda\lambda}^2(\omega) + \epsilon_{2,\lambda\lambda}^2(\omega)} - \epsilon_{1,\lambda\lambda}(\omega)\right]^{1/2} \tag{8}$$

$$n_\lambda(\omega) = \left[\frac{\sqrt{\epsilon_{1,\lambda\lambda}^2(\omega) + \epsilon_{2,\lambda\lambda}^2(\omega)} + \epsilon_{1,\lambda\lambda}(\omega)}{2}\right]^{1/2} \tag{9}$$

$$k_\lambda(\omega) = \left[\frac{\sqrt{\epsilon_{1,\lambda\lambda}^2(\omega) + \epsilon_{2,\lambda\lambda}^2(\omega)} - \epsilon_{1,\lambda\lambda}(\omega)}{2}\right]^{1/2} \tag{10}$$

$$R_\lambda(\omega) = \frac{[n_\lambda(\omega) - 1]^2 + k_\lambda^2(\omega)}{[n_\lambda(\omega) + 1]^2 + k_\lambda^2(\omega)} \tag{11}$$

$$L_\lambda(\omega) = \mathrm{Im}\left[-\frac{1}{\epsilon_{\lambda\lambda}(\omega)}\right] = \frac{\epsilon_{2,\lambda\lambda}(\omega)}{\epsilon_{1,\lambda\lambda}^2(\omega) + \epsilon_{2,\lambda\lambda}^2(\omega)} \tag{12}$$

here, c is the speed of light in vacuum.

For a two-dimensional structure represented by a periodically repeated supercell, the calculated dielectric tensor contains the combined response of the material and the vacuum region. We therefore restore the intrinsic dielectric function of each two-dimensional structure using the dielectric-function renormalization method of Yang and Gao [32]:

$$(\epsilon_2)_{2D} = \frac{d_{SC}}{d_{2D}}(\epsilon_2)_{SC} \tag{13}$$

$$(\epsilon_1)_{2D} = 1 + \frac{d_{SC}}{d_{2D}}\left[(\epsilon_1)_{SC} - 1\right] \tag{14}$$

here, $(\epsilon_i)_{2D}$ and $(\epsilon_i)_{SC}$, with $i = 1, 2$, are the intrinsic dielectric function of the two-dimensional structure and the dielectric function of the full supercell, respectively; d_{SC} is the length of the supercell normal to the two-dimensional structure; and d_{2D} is the thickness assigned to the structure. The values of d_{2D} and the values of `NBANDS` used in the dielectric function calculations are provided in the Supplementary Information.

III. RESULTS AND DISCUSSION

A. Atomic structure

We first examine the atomic structures of the bulk reference phases, pristine two-dimensional beryllium phases, and hydrogen-functionalized beryllene structures considered in this work. The bulk reference structures are shown in figure 1, whereas the pristine two-dimensional structures are shown in figure 2. The hydrogen-functionalized structures are shown in figure 3.

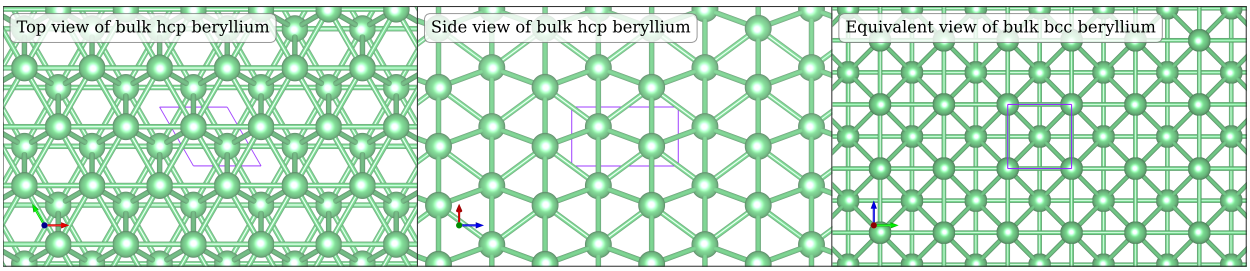

FIG. 1. Atomic structures of the bulk beryllium phases. Left and middle: top and side views of bulk hcp beryllium; Right: equivalent view of bulk bcc beryllium.

Bulk beryllium is known experimentally to adopt the hexagonal close-packed structure under ambient conditions, whereas the body-centred cubic phase has been predicted at high pressure but has not been conclusively observed experimentally. We include this phase as a bulk reference structure. The optimized cohesive energy of bulk hcp beryllium is $-3.73\,\mathrm{eV}$

per atom, which is 0.10 eV per atom lower than that of bulk bcc beryllium. These two bulk phases provide useful reference points for comparing the bonding geometry, electronic structure, and optical response of the low-dimensional beryllium systems discussed below.

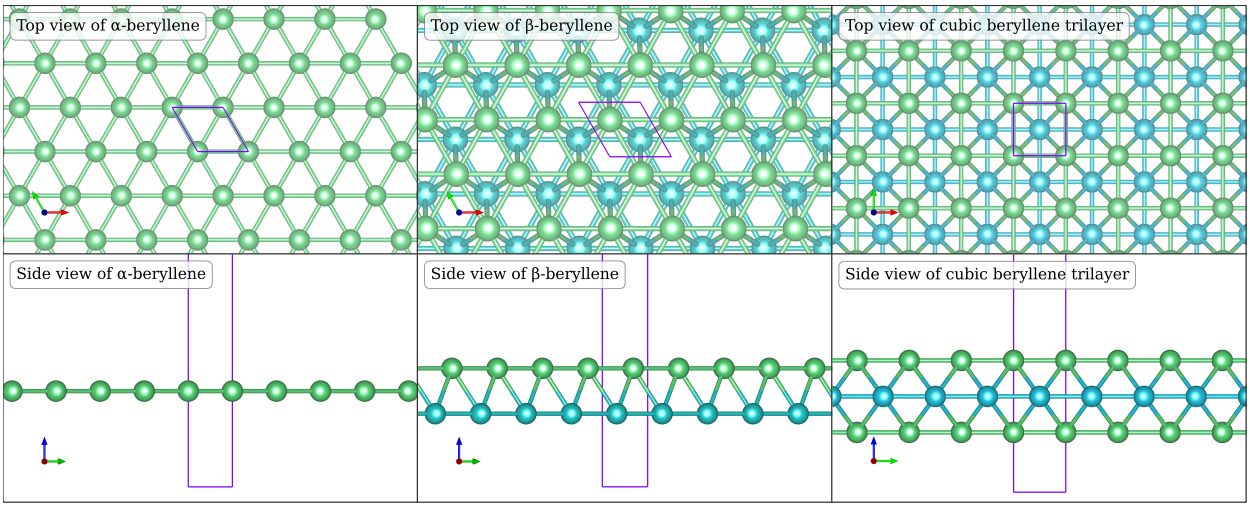

FIG. 2. Atomic structures of the pristine two-dimensional beryllium phases. Left: top and side views of α-beryllene; Middle: top and side views of β-beryllene; Right: top and side views of cubic trilayer beryllene.

Among the pristine two-dimensional phases, α-beryllene and β-beryllene correspond to the previously proposed hexagonal monolayer and bilayer forms of beryllene. α-beryllene forms a planar hexagonal network, where each beryllium atom bonds to six neighbouring beryllium atoms. β-beryllene can be regarded as a bilayer form of α-beryllene, where beryllium atoms in the upper layer sit above hollow sites of the lower layer. In addition to these known phases, we also identify a cubic trilayer beryllene structure. This structure adopts an AB stacking arrangement, where the beryllium atoms in the central layer are located at the hollow sites of the upper and lower layers. It therefore provides a distinct structural motif within the same elemental system.

The cohesive energies of α-beryllene, β-beryllene, and cubic trilayer beryllene are 0.79 eV, 0.44 eV, and 0.43 eV per atom less favourable than that of bulk hcp beryllium, respectively. Compared with bulk hcp beryllium, the in-plane lattice constants of α-beryllene and β-beryllene are reduced by 6.0 % and 6.4 %, respectively. The interlayer beryllium–beryllium distance in β-beryllene is 6.5 % larger than that in bulk hcp beryllium. For cubic trilayer beryllene, the in-plane lattice constant is 0.8 % larger than that of bulk bcc beryllium,

whereas the interlayer distance is reduced by 17 %.

We also tested two-layer and four-layer cubic beryllene structures, as well as graphene-like beryllium layers from one to four layers. These structures were found to be dynamically unstable from the phonon dispersion calculations, and are therefore not discussed further.

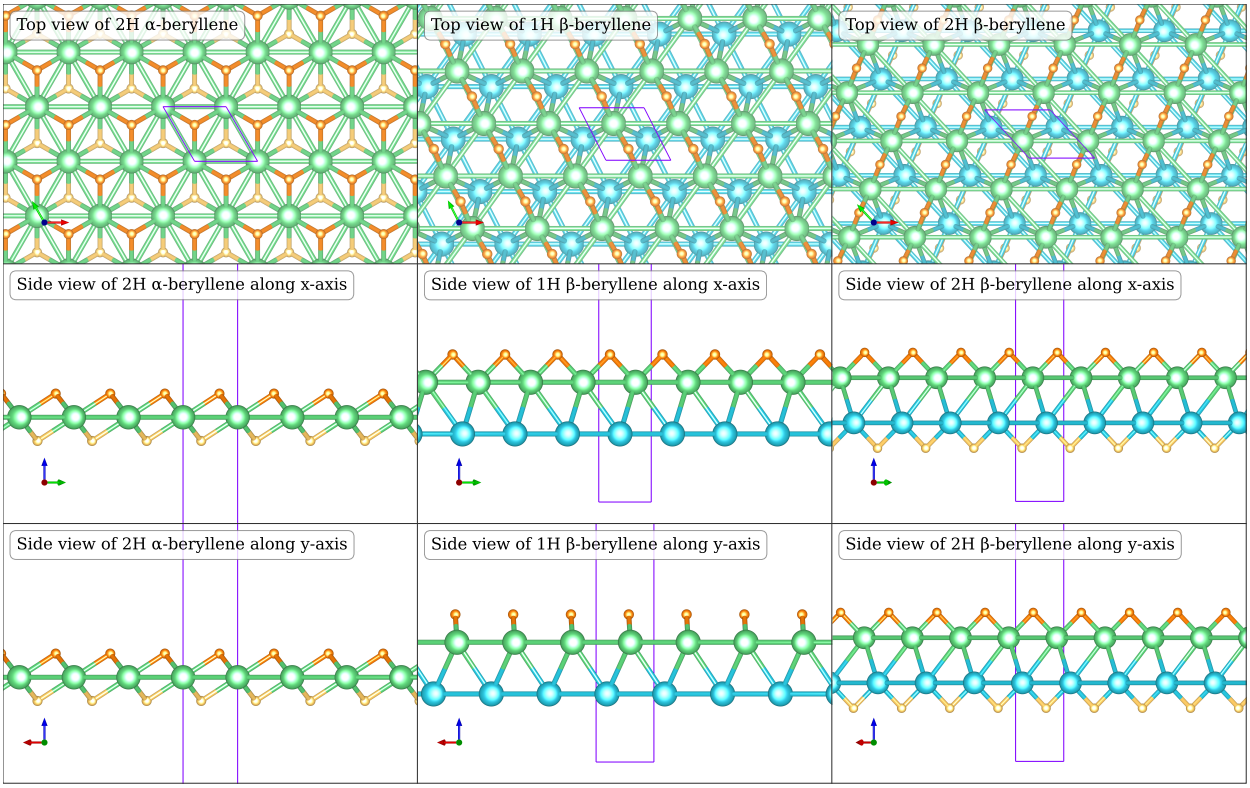

FIG. 3. Atomic structures of the hydrogen-functionalized beryllene phases. Left: double-sided hydrogen-terminated α-beryllene; Middle: single-sided hydrogen-terminated β-beryllene; Right: double-sided hydrogen-terminated β-beryllene. Upper row: top views; Middle and lower rows: side views along the x and y axes, respectively.

We then consider hydrogen functionalization of the two-dimensional phases. For α-beryllene, we consider the double-sided hydrogen-terminated configuration. For β-beryllene, both single-sided and double-sided hydrogen-terminated configurations are considered. These hydrogen-functionalized structures preserve the underlying beryllium framework but modify the local bonding environment around the surface beryllium atoms. As a result, hydrogen adsorption is expected to affect the energetic stability, electronic structure, and optical response of the corresponding beryllene phases.

The optimized structural parameters, cohesive energies, and adsorption energies provide

the structural basis for the dynamical, electronic, topological, and optical analyses below.

B. Dynamical stability and phonon dispersion

We then evaluate the dynamical stability of the beryllium structures using phonon dispersion calculations. The phonon dispersion curves are calculated using finite differences with 5×5 supercells. The bulk hcp and bulk bcc structures are dynamically stable, as are the pristine α-beryllene and β-beryllene structures, as shown in figure S13 of the Supplementary Information. This result is consistent with previous reports on α- and β-beryllene.

Hydrogen functionalization can modify the structural and electronic properties of a two-dimensional material by changing the local bonding environment. We therefore examine hydrogen adsorption on the beryllene structures. Among the hydrogen adsorption configurations considered, the structures that satisfy both energetic and dynamical stability criteria are double-sided hydrogen-terminated α-beryllene, single-sided hydrogen-terminated β-beryllene, and double-sided hydrogen-terminated β-beryllene. For brevity, we denote these structures as 2H-α-beryllene, 1H-β-beryllene, and 2H-β-beryllene, respectively.

The phonon dispersion curves of cubic trilayer beryllene and the hydrogen-functionalized structures are shown in figure 4. No significant imaginary frequencies appear in these spectra, which indicates that cubic trilayer beryllene, 2H-α-beryllene, 1H-β-beryllene, and 2H-β-beryllene are dynamically stable. The Brillouin zones used for these phonon paths are shown in figure S14 of the Supplementary Information.

We further examine the finite-temperature stability of the hydrogen-functionalized structures using *ab initio* molecular dynamics simulations. The AIMD trajectories and the corresponding atomic structures are shown in figure 5. For all three hydrogen-functionalized structures, the total energy fluctuates around a stable average value, and the temperature remains close to the target range during the simulation. The corresponding atomic snapshots preserve the beryllene frameworks without structural collapse. These results support the finite-temperature stability of the hydrogen-functionalized phases.

The optimized structural parameters of the hydrogen-functionalized structures are listed in table I. For 2H-α-beryllene, the in-plane lattice constant increases from 2.129 Å in pristine α-beryllene to 2.322 Å after hydrogen adsorption. For 1H-β-beryllene, hydrogen adsorption breaks the hexagonal symmetry: the two in-plane lattice constants become 2.322 Å and

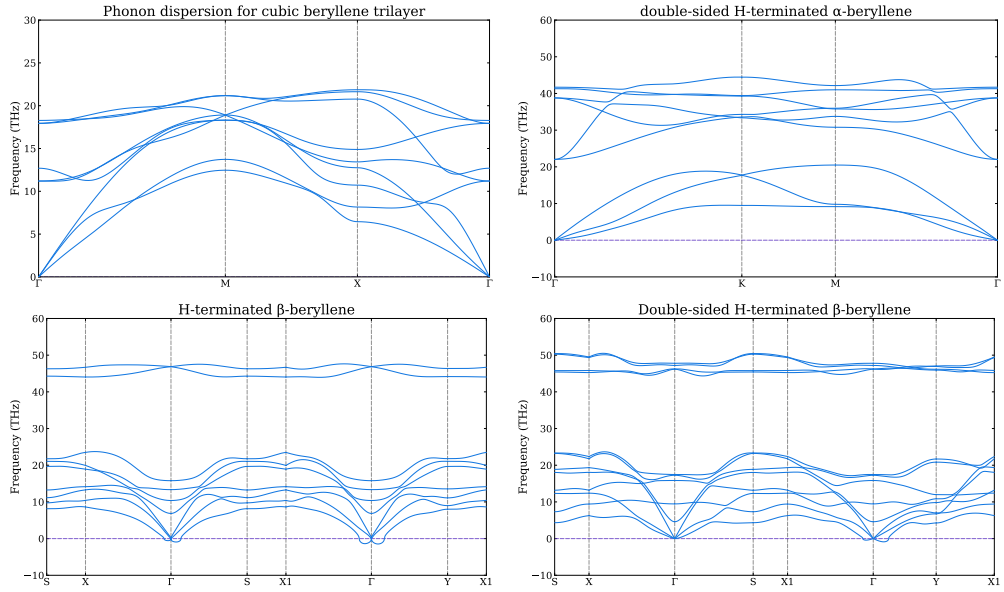

FIG. 4. Phonon dispersion curves of cubic trilayer beryllene and the hydrogen-functionalized beryllene structures. Upper left: cubic trilayer beryllene; Upper right: 2H-α-beryllene; Lower left: 1H-β-beryllene; Lower right: 2H-β-beryllene.

2.099 Å, with an angle of 117° between the lattice vectors. For 2H-β-beryllene, the two lattice vectors have the same magnitude of 2.590 Å, with an angle of 133°. This geometry forms a rhombic lattice. The hydrogen adsorption energies are −3.70 eV, −3.46 eV, and −3.55 eV for 2H-α-beryllene, 1H-β-beryllene, and 2H-β-beryllene, respectively. These negative adsorption energies indicate that all three hydrogen-functionalized structures are stable with respect to molecular hydrogen.

system	a	b	interlayer Be-Be (Å)	H-Be (Å)	E_{ad} (eV)	angle	electronic nature
2H-α-beryllene	2.322	2.322	–	1.599	−3.70	120°	semiconductor
1H-β-beryllene	2.322	2.099	1.849	1.450	−3.46	117°	metallic
2H-β-beryllene	2.590	2.590	1.786	1.455	−3.55	133°	metallic

TABLE I. Lattice vectors a and b, interlayer Be-Be distance, H-Be bond length, hydrogen adsorption energy E_{ad}, angle between the two in-plane lattice vectors, and electronic nature of the hydrogen-functionalized beryllene structures. The lattice vectors, interlayer Be-Be distance, and H-Be bond length are given in Å, while the adsorption energy is given in eV.

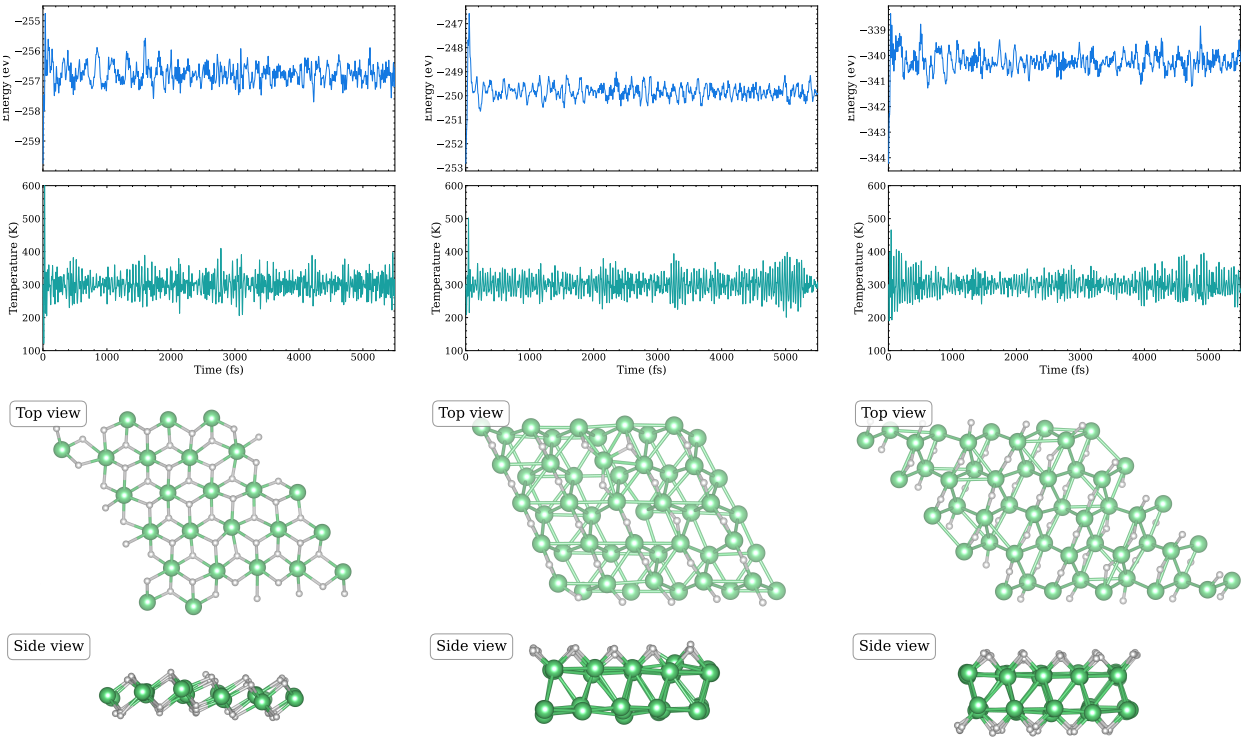

FIG. 5. *Ab initio* molecular dynamics simulations and corresponding atomic structures of the hydrogen-functionalized beryllene phases. Left: 2H-α-beryllene; Middle: 1H-β-beryllene; Right: 2H-β-beryllene. The upper row shows the total energy and temperature as functions of simulation time, and the lower row shows the corresponding top and side views of the atomic structures.

C. Electronic properties

We now examine the electronic properties of pristine and hydrogen-functionalized beryllene structures. The band structures of pristine α-beryllene and β-beryllene are shown in figure S15 of the Supplementary Information, and agree well with previous calculations [10]. The band structures of cubic trilayer beryllene and the hydrogen-functionalized structures are shown in figure 6.

The cubic trilayer beryllene structure shows metallic behaviour, with multiple bands crossing the Fermi level. The bands near the Fermi level show moderate dispersion, which suggests delocalized electronic states and good in-plane conductivity. Around the Γ, M, and X points, noticeable band splitting appears, consistent with interlayer coupling in the trilayer structure. The coexistence of relatively flat bands and dispersive unoccupied bands at higher energies further indicates a mixture of localized and itinerant electronic states.

Hydrogen functionalization changes the electronic structure differently for the α and β phases. For 2H-α-beryllene, the band structure shows a wide indirect band gap exceeding 4.0 eV at the PBE level. Since PBE generally underestimates band gaps, the actual gap is expected to be larger. In contrast, both 1H-β-beryllene and 2H-β-beryllene remain metallic, with bands crossing the Fermi level. These results show that hydrogen functionalization opens a large gap in α-beryllene, whereas the hydrogen-functionalized β-based structures preserve their metallic character.

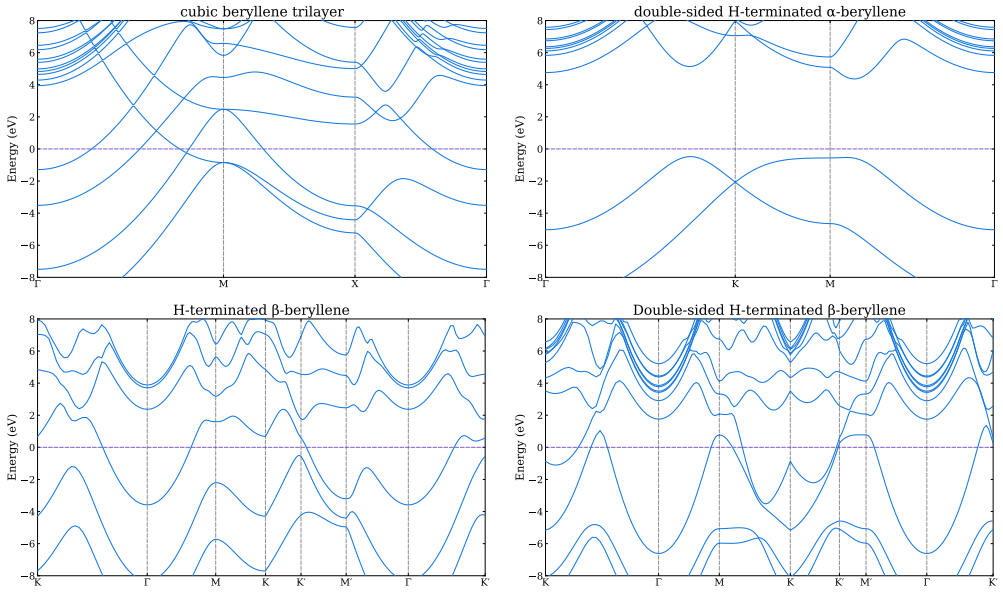

FIG. 6. Band structures of cubic trilayer beryllene and the hydrogen-functionalized beryllene structures. Upper left: cubic trilayer beryllene; Upper right: 2H-α-beryllene; Lower left: 1H-β-beryllene; Lower right: 2H-β-beryllene.

The density of states further supports the band-structure interpretation. Figure 7 compares the total density of states of bulk bcc beryllium and cubic trilayer beryllene. The cubic trilayer structure has a finite density of states at the Fermi level, consistent with its metallic band structure.

The total and projected densities of states of pristine and hydrogen-functionalized beryllene structures are shown in figure 8. For 2H-α-beryllene, the density of states vanishes around the Fermi level, consistent with the wide-gap semiconducting band structure. For 1H-β-beryllene and 2H-β-beryllene, the finite density of states at the Fermi level confirms their metallic behaviour. The projected densities of states show strong hybridization among beryllium s and p states, together with the hydrogen 1s state in the hydrogen-functionalized

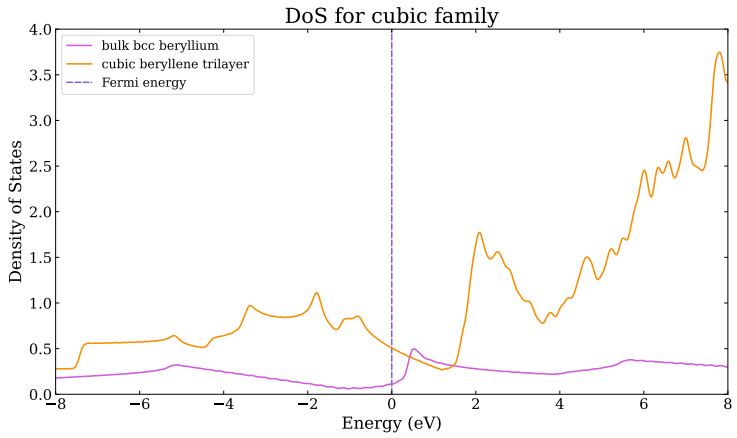

FIG. 7. Total density of states of bulk bcc beryllium and cubic trilayer beryllene.

structures. This hybridization reflects the modification of the local bonding environment after hydrogen functionalization.

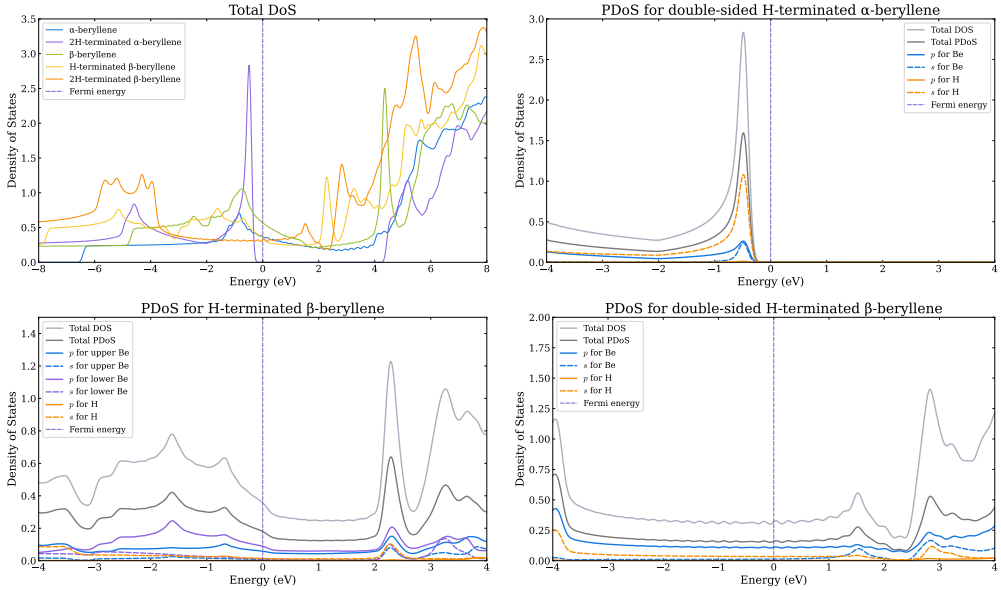

FIG. 8. Total and projected densities of states for pristine and hydrogen-functionalized beryllene structures. Upper left: total density of states; Upper right: projected density of states of 2H-α-beryllene; Lower left: projected density of states of 1H-β-beryllene; Lower right: projected density of states of 2H-β-beryllene.

D. Topological properties

Previous studies reported non-trivial topology in α-beryllene, whereas β-beryllene was classified as topologically trivial [10]. Since these beryllene structures have inversion symmetry, we verify their topological character using the Z_2 invariant calculated from the parity eigenvalues of the occupied spin-orbit-coupled states at the time-reversal invariant momenta. We also apply the same analysis to cubic trilayer beryllene.

For these calculations, we use an energy cutoff of 520 eV, a $24 \times 24 \times 1$ k-point mesh, and a vacuum region of 15 Å. The self-consistent convergence criterion is set to 1×10^{-7} eV, and Gaussian smearing of 0.02 eV is used. Spin-orbit coupling is included in the band-structure and parity calculations. The parity eigenvalues of the spinor wave functions are calculated using `irvsp` [27].

For a time-reversal invariant momentum Λ_i, the parity product is given by:

$$\delta_i = \prod_{m=1}^{N_{\mathrm{occ}}/2} \xi_{2m}(\Lambda_i) \tag{15}$$

here, $\xi_{2m}(\Lambda_i) = \pm 1$ is the parity eigenvalue of one state in the mth Kramers pair at Λ_i. The Z_2 index is then obtained from the Fu–Kane product [28, 29]:

$$(-1)^{\nu} = \prod_i \delta_i \tag{16}$$

here, $\nu = 1$ denotes a non-trivial topology, whereas $\nu = 0$ denotes a trivial topology.

The calculated parity eigenvalues, parity products, Z_2 indices, and topological classifications are listed in table II. For α-beryllene and β-beryllene, the three M points are equivalent by symmetry, so the Fu–Kane product reduces to $(-1)^{\nu} = \delta_\Gamma(\delta_\mathrm{M})^3 = \delta_\Gamma\delta_\mathrm{M}$. For cubic trilayer beryllene, the two X points contribute as $(\delta_\mathrm{X})^2$, and the product reduces to $(-1)^{\nu} = \delta_\Gamma\delta_\mathrm{M}(\delta_\mathrm{X})^2 = \delta_\Gamma\delta_\mathrm{M}$.

For α-beryllene, the parity products at the Γ and M points have opposite signs. The Fu–Kane product is therefore $(-1)^{\nu} = -1$, which gives $\nu = 1$ and confirms the non-trivial topology previously reported for this phase. For β-beryllene, the parity products give $(-1)^{\nu} = +1$, which gives $\nu = 0$ and confirms that this phase is topologically trivial. For cubic trilayer beryllene, the parity products again change sign between the Γ and M points, giving $(-1)^{\nu} = -1$ and $\nu = 1$. Therefore, cubic trilayer beryllene is also a non-trivial Z_2

structure	Γ	M	product	ν	classification
α-beryllene	$+,+$	$+,-$	-1	1	non-trivial
β-beryllene	$+,-,+,-$	$-,+,+,-$	$+1$	0	trivial
cubic trilayer beryllene	$-,+,+,+,-,+$	$-,+,-,+,-,+$	-1	1	non-trivial

TABLE II. Parity eigenvalues at the time-reversal invariant momenta, parity products, Z_2 indices, and topological classifications for α-beryllene, β-beryllene, and cubic trilayer beryllene.

topological phase.

E. Optical properties

We now examine the optical properties of the pristine and hydrogen-functionalized beryllene structures, together with the bulk hcp and bulk bcc beryllium reference phases. The dielectric function, absorption coefficient, refractive index, extinction coefficient, reflectivity, and energy-loss spectrum are defined in section II. Here, we use these quantities to compare the direction-dependent optical response of the beryllene structures.

1. Pristine α- and β-beryllene

We first discuss the optical response of pristine α-beryllene and β-beryllene, which then provides a reference for the bulk, cubic trilayer, and hydrogen-functionalized structures. The real and imaginary parts of the dielectric function for the in-plane and out-of-plane directions are shown in figure 9.

For β-beryllene, the in-plane real part indicates two plasmon excitations at approximately 2.5 eV and 5.3 eV, as well as plasmon-like features above approximately 7 eV. These features occur together with relatively small values of the imaginary part. In the out-of-plane direction, the low-energy plasmon feature is absent. For α-beryllene, the response differs, and only a higher-energy plasmon-like feature appears around 7 eV to 9 eV. Interband transitions may coexist with these plasmonic features and form a broad background. When interband transitions and plasmons occupy the same energy range, they may couple and produce finite plasmon lifetimes together with interband-transition character.

The real part also shows that β-beryllene behaves as an indefinite medium over part of the low-energy range. In the region of approximately 1.2 eV to 2.2 eV, the in-plane and out-of-plane components of the real part have opposite signs. This behaviour signals direction-dependent plasmonic modes.

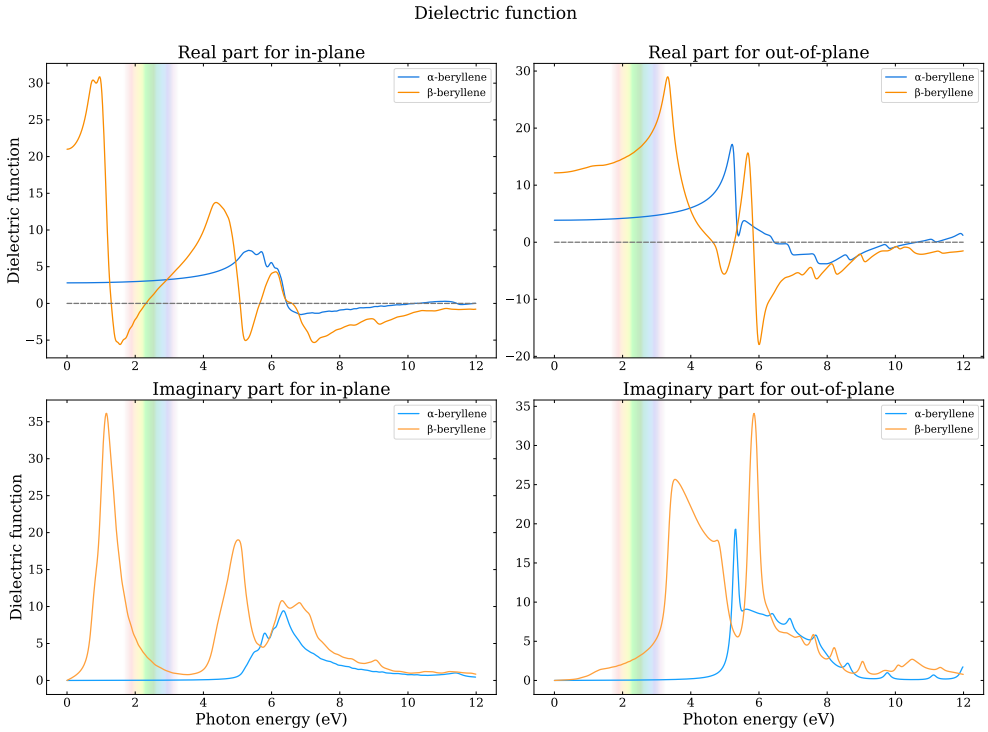

FIG. 9. Real and imaginary parts of the dielectric function for α-beryllene and β-beryllene. The spectra are shown for the in-plane and out-of-plane directions as functions of photon energy.

The absorption coefficient and refractive index are shown in figure 10. The absorption spectra show differences between the two optical directions and between the two structures. In the in-plane channel, β-beryllene has a stronger absorption response than α-beryllene over a broad photon-energy range, including a noticeable peak in the visible-light region. In contrast, α-beryllene develops its main absorption only beyond the low-energy range and then forms a broader spectral feature. In the out-of-plane channel, both structures develop stronger peak structures only beyond the visible range, while β-beryllene still shows the stronger overall absorption. These trends show that the optical attenuation depends on both the optical direction and the crystal structure.

The refractive index spectra follow the same structural and directional contrast. In the in-plane channel, β-beryllene develops two pronounced enhancements in the infrared and near-

ultraviolet regions, with a minimum in the visible range. By contrast, α-beryllene varies more smoothly over the same energy interval. In the out-of-plane channel, β-beryllene remains stronger than α-beryllene over the infrared to near-ultraviolet range. For both structures, the refractive index drops below one above approximately 8 eV, which occurs close to the negative region of the real part of the dielectric function. This behaviour is consistent with the strong interaction between light and plasmonic oscillations near the plasmon resonance.

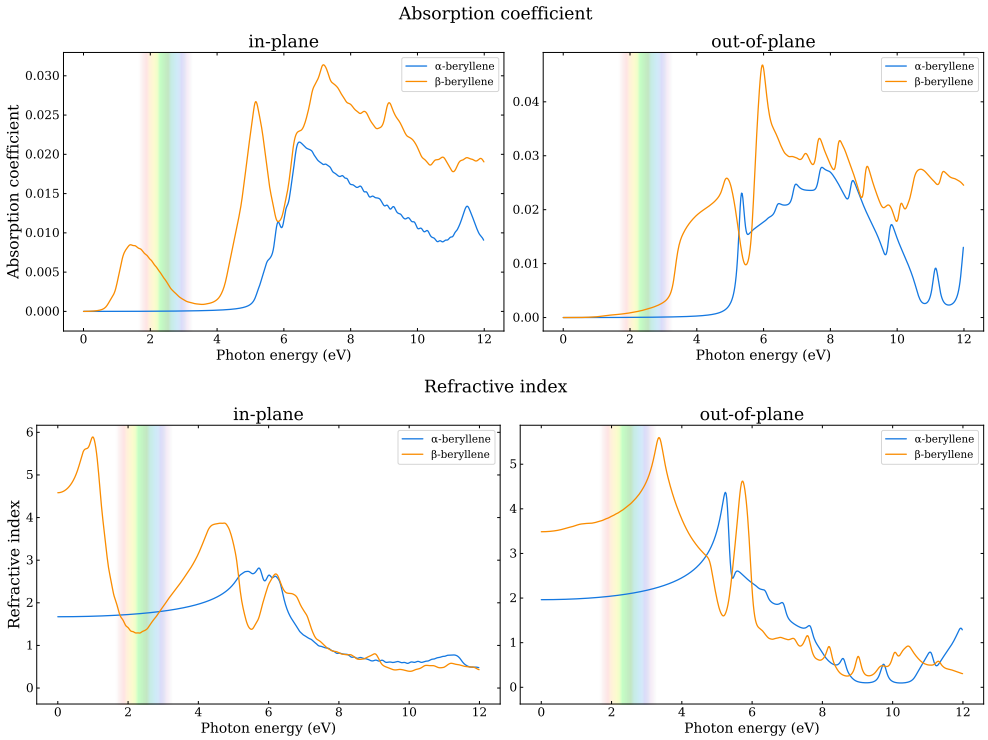

FIG. 10. Absorption coefficient and refractive index of α-beryllene and β-beryllene. Upper: absorption coefficient; Lower: refractive index. The spectra are shown for the in-plane and out-of-plane directions as functions of photon energy.

The extinction coefficient is shown in figure 11. In the in-plane channel, β-beryllene develops strong peaks in the infrared and near-ultraviolet regions, separated by a pronounced trough. By contrast, α-beryllene again varies more smoothly. In the out-of-plane channel, β-beryllene remains stronger over the low-energy range and then decreases rapidly toward the near-ultraviolet region. This behaviour shows that amplitude attenuation also depends on both optical direction and crystal structure.

The corresponding reflectivity and energy-loss spectra show the same structural and directional contrast. For reflectivity, β-beryllene shows strong reflection in the low-energy

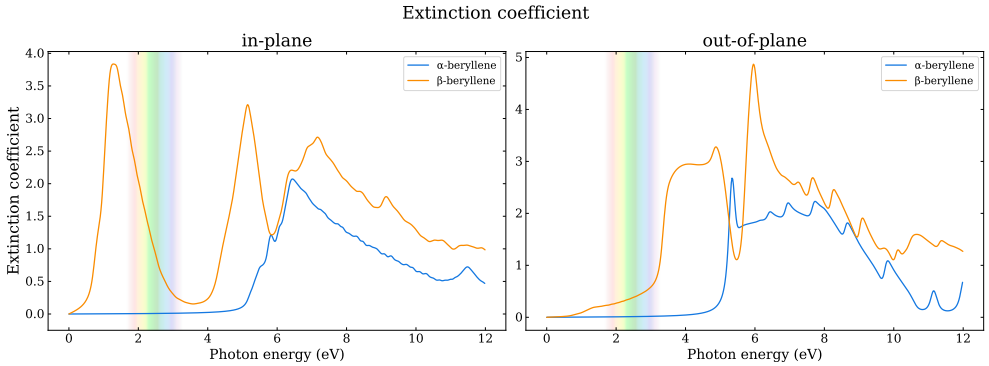

FIG. 11. Extinction coefficient of α-beryllene and β-beryllene for the in-plane and out-of-plane directions as a function of photon energy.

and near-ultraviolet regions for in-plane polarization, separated by a pronounced minimum. In contrast, α-beryllene changes more smoothly and shows a broad intermediate-energy feature. For out-of-plane polarization, β-beryllene remains stronger over the low-energy range, whereas α-beryllene shows pronounced peaks in the ultraviolet region.

The energy-loss spectrum of β-beryllene shows pronounced peaks in the visible and deep-ultraviolet regions in the in-plane channel, with the visible-range peak being stronger. By contrast, α-beryllene remains weak at low photon energy and then rises gradually on entering the deep-ultraviolet region. In the out-of-plane channel, both structures remain nearly zero in the visible range and then increase toward the near-ultraviolet region. These spectra identify the dominant dissipation channels associated with the inverse dielectric response.

2. Bulk hcp, bulk bcc, and cubic trilayer beryllene

We next compare the optical response of bulk hcp beryllium, bulk bcc beryllium, and cubic trilayer beryllene. The real and imaginary parts of the dielectric function are shown in figure 12. The bulk hcp and bulk bcc structures show similar behaviour and weak directional dependence. For both bulk structures, the real part becomes negative above approximately 4.5 eV, reaches a minimum around 5.5 eV to 6.0 eV, and then rises smoothly toward zero around 12 eV. The imaginary part displays a broad low-energy feature up to approximately 5.5 eV, indicating a wide range of allowed interband transitions. It then decreases sharply around 6 eV, coinciding with the minimum in the real part and indicating a plasmonic optical response.

For cubic trilayer beryllene, the most prominent feature is its enhanced visible-region optical response compared with bulk bcc beryllium. A pronounced feature appears around 2.0 eV to 2.5 eV, together with a redistribution of spectral weight toward lower excitation energies. This indicates increased electronic polarizability in the trilayer. In the out-of-plane direction, this low-energy peak becomes even more pronounced. The rapid decrease of the real part to negative values above approximately 3.8 eV indicates the onset of metallic-like screening and plasmonic behaviour, with a negative region extending over a broader energy range than in bulk beryllium. The deeper minimum around 5.5 eV to 6.0 eV and the stronger imaginary dielectric peak further suggest an increased density of optically active electronic states arising from quantum confinement and surface-derived bands. The large peak of the imaginary part near 3.7 eV to 4.0 eV indicates strong interband transitions in the trilayer, whereas in the out-of-plane direction the corresponding large peak appears at a lower energy near 2.5 eV.

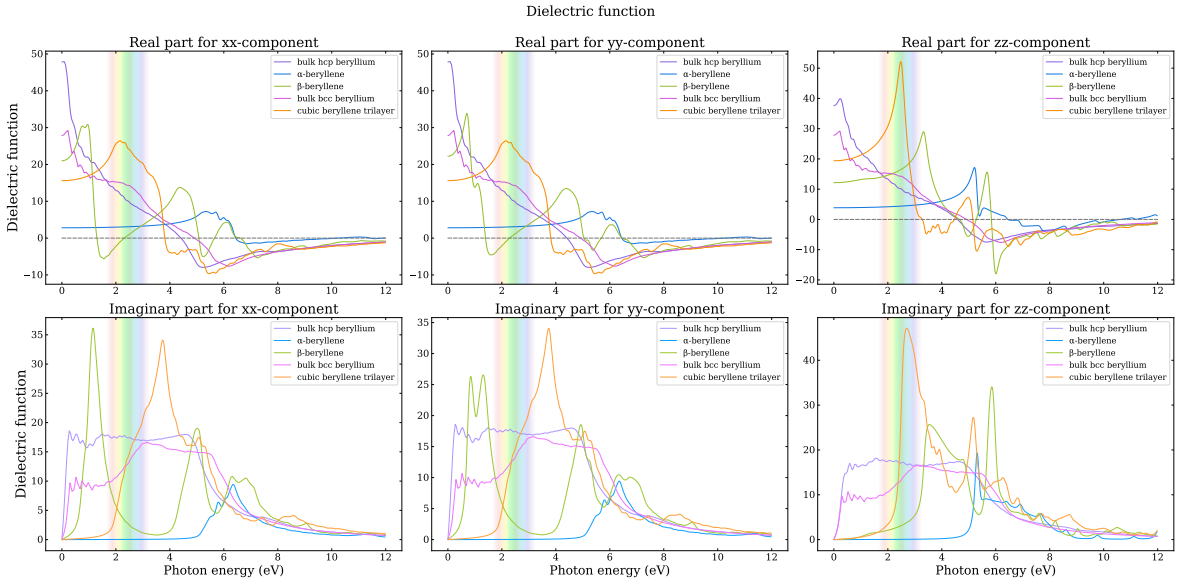

FIG. 12. Real and imaginary parts of the dielectric function for bulk hcp beryllium, bulk bcc beryllium, α-beryllene, β-beryllene, and cubic trilayer beryllene. The spectra are shown for the xx, yy, and zz components as functions of photon energy.

The absorption coefficient and energy-loss spectra are shown in figure 13. The absorption spectra show that reducing cubic beryllium to cubic trilayer beryllene enhances the optical response. Compared with the bulk phase, the trilayer exhibits an earlier absorption onset

and larger absorption coefficients throughout the visible and near-ultraviolet regions, with a broad maximum near 5.5 eV to 6.0 eV. The enhanced intensity and slight redshift of the main absorption features indicate stronger optical transitions caused by the modified electronic structure of the low-dimensional system. The trilayer also displays several pronounced absorption maxima between 5 eV to 9 eV, reflecting a richer distribution of interband transitions than in the bulk material. These results suggest that quantum confinement and surface-induced electronic states increase the optical activity of cubic trilayer beryllene over a broad spectral range.

The energy-loss spectra show that, unlike β-beryllene with its pronounced low-energy plasmonic peak near 2 eV, the other structures remain almost featureless below approximately 6 eV. At higher energies, broad loss features begin to develop, especially around and above 6 eV. These features are consistent with the onset of interband transitions and collective electronic response.

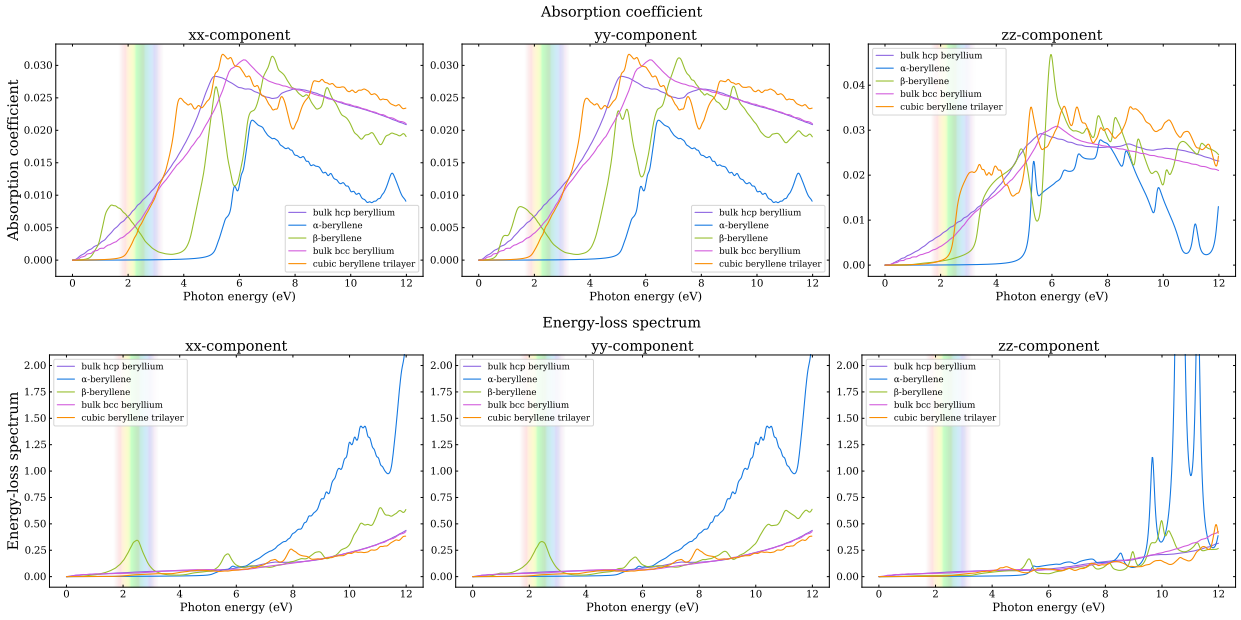

FIG. 13. Absorption coefficient and energy-loss spectrum for bulk hcp beryllium, bulk bcc beryllium, α-beryllene, β-beryllene, and cubic trilayer beryllene. Upper: absorption coefficient; Lower: energy-loss spectrum. The spectra are shown for the xx, yy, and zz components as functions of photon energy.

3. Hydrogen-functionalized beryllene structures

We now examine the effect of hydrogen adsorption on the optical properties of α-beryllene and β-beryllene. The real and imaginary parts of the dielectric function for the pristine and hydrogen-functionalized structures are shown in figure 14.

For α-beryllene, the pristine structure exhibits a relatively low and slowly varying real part in the low-energy region below approximately 4 eV, followed by a modest peak near 5 eV to 6 eV. In contrast, 2H-α-beryllene shows a strongly enhanced dielectric response, with values reaching approximately 60–70 near 5 eV. This increase indicates a strong enhancement of electronic polarizability after hydrogen adsorption. The hydrogen-functionalized structure also shows a sharp transition from large positive values to negative values around 5.5 eV to 6.0 eV, indicating plasma-like behaviour and a shifted zero crossing relative to pristine α-beryllene. This shift suggests that hydrogen functionalization modifies the electronic structure by introducing new hybridized states and changing the band dispersion, thereby strengthening collective electronic oscillations.

For the imaginary part, the difference is also pronounced. Pristine α-beryllene shows relatively weak absorption with a modest peak centred around 6 eV. By contrast, 2H-α-beryllene displays an intense and broadened absorption peak near 5.0 eV to 5.5 eV, together with additional features at higher photon energies. This enhancement and slight redshift of the dominant peak indicate that hydrogen adsorption introduces or strengthens optically allowed transitions, likely through hybridization between hydrogen states and the host electronic states. The broadening also suggests an increased density of accessible electronic transitions over a wider energy range. Consequently, 2H-α-beryllene is expected to exhibit substantially stronger optical absorption in the visible to near-ultraviolet region than pristine α-beryllene.

For β-beryllene, the pristine phase shows a moderate low-energy dielectric response, followed by a rapid decrease into negative values around 1 eV to 2 eV, and then a recovery with a broad positive feature near 4 eV to 5 eV. Upon single-sided hydrogen adsorption, the magnitude of the real part is slightly enhanced in the low-energy range, and the spectral features become shifted and smoothed. The zero crossing shifts to higher energies, around 3 eV to 4 eV, and the oscillatory behaviour becomes less abrupt. This behaviour indicates a redistribution of electronic states and a reduction of sharp interband features. For double-sided

hydrogen adsorption, these effects become stronger. The real part increases over a wider energy range and shows a more gradual dispersion, with a delayed and deeper transition into negative values near 6 eV to 6.5 eV. This progression suggests that increasing hydrogen coverage enhances electronic screening and modifies the effective plasma frequency, consistent with stronger perturbation of the band structure and increased carrier polarization.

For the imaginary part of β-beryllene, the pristine phase shows a sharp and intense low-energy absorption peak near 1 eV to 1.5 eV, followed by a weaker and broader feature near 5 eV. For 1H-β-beryllene, the low-energy peak is reduced in intensity and broadened, while a more prominent absorption feature emerges around 3.5 eV to 4.0 eV. This indicates a shift of the dominant interband transitions toward higher energies. For 2H-β-beryllene, the spectral weight is redistributed even more strongly: the low-energy peak nearly vanishes, and a strong broad absorption peak appears around 6 eV to 6.5 eV. This substantial blue shift and enhancement suggest that hydrogen adsorption suppresses low-energy transitions and promotes higher-energy electronic excitations, likely through band-structure modification and the formation of new hybridized states involving hydrogen.

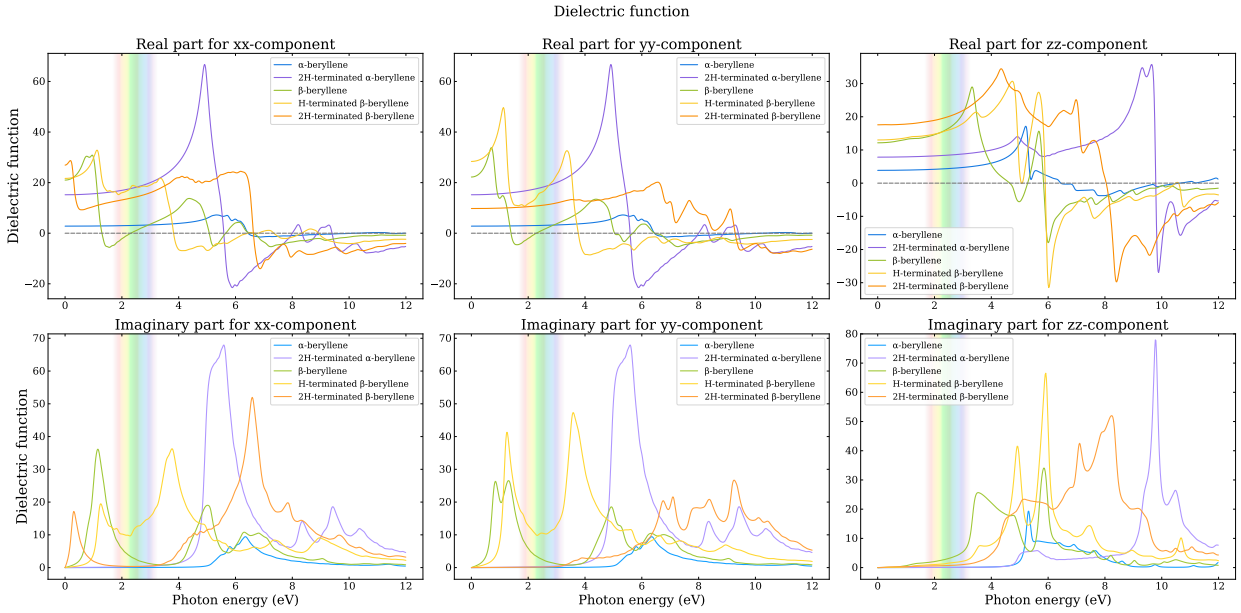

FIG. 14. Real and imaginary parts of the dielectric function for pristine and hydrogen-functionalized α- and β-beryllene structures. The spectra are shown for the xx, yy, and zz components as functions of photon energy.

The corresponding absorption coefficient and energy-loss spectra are shown in figure 15.

For α-beryllene, hydrogen adsorption strongly enhances the absorption response in the visible to near-ultraviolet range. For β-beryllene, hydrogen adsorption shifts spectral weight from the low-energy region toward higher photon energies. The progression from pristine β-beryllene to 1H-β-beryllene and then to 2H-β-beryllene therefore shows a systematic redistribution of optical spectral weight with increasing hydrogen coverage. The corresponding energy-loss spectra follow the same trend, indicating that hydrogen functionalization modifies both interband transitions and collective electronic excitations.

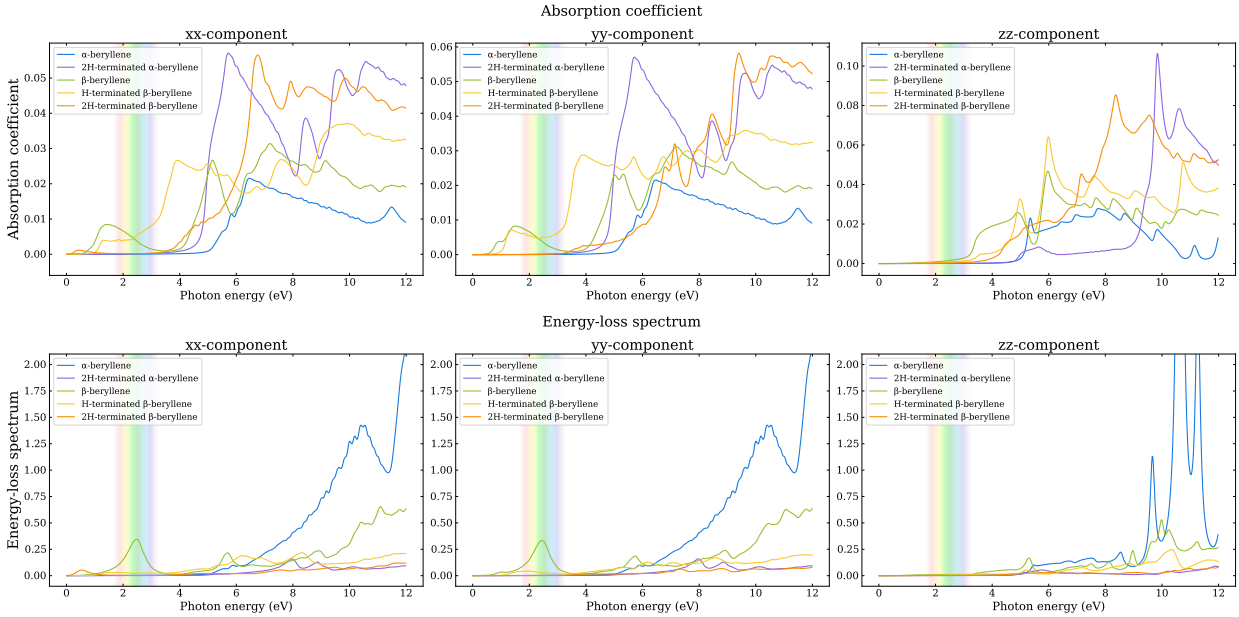

FIG. 15. Absorption coefficient and energy-loss spectrum for pristine and hydrogen-functionalized α- and β-beryllene structures. Upper: absorption coefficient; Lower: energy-loss spectrum. The spectra are shown for the xx, yy, and zz components as functions of photon energy.

These results show that both dimensionality and hydrogen functionalization strongly modify the optical response of beryllene. The pristine α and β structures already display pronounced anisotropy and plasmon-related features. The cubic trilayer shifts optical spectral weight toward lower energies and enhances visible-to-near-ultraviolet absorption. Hydrogen adsorption further redistributes the dielectric response and changes the characteristic excitation energies through modified electronic hybridization and screening.

IV. CONCLUSION

In this work, we investigated the structural stability, electronic structure, optical response, and topological properties of pristine and hydrogen-functionalized two-dimensional beryllium phases using first-principles density functional theory calculations. Alongside the previously reported α- and β-beryllene structures, we identified a dynamically stable cubic trilayer beryllene phase. The phonon dispersion calculations also confirm the dynamical stability of 2H-α-beryllene, 1H-β-beryllene, and 2H-β-beryllene, while the molecular dynamics simulations further support the finite-temperature stability of these hydrogen-functionalized structures.

Hydrogen functionalization substantially modifies the structural and electronic properties of beryllene. The calculated adsorption energies indicate that the hydrogen-functionalized structures are stable with respect to molecular hydrogen. In particular, 2H-α-beryllene transforms from a metallic system into a wide-gap semiconductor, whereas the hydrogen-functionalized β-based structures retain metallic behaviour.

The calculated optical properties demonstrate that dimensional reduction and hydrogen functionalization strongly modify the dielectric response of beryllium. Compared with the bulk reference phases, cubic trilayer beryllene exhibits increased spectral weight at lower excitation energies and enhanced absorption in the visible-to-near-ultraviolet region. Pristine α- and β-beryllene show pronounced optical anisotropy and plasmon-related features. Hydrogen functionalization further redistributes the optical spectral weight and shifts the characteristic excitation energies through changes in electronic hybridization and screening.

Parity analysis of the spin-orbit-coupled electronic states confirms the previously reported non-trivial topology of α-beryllene and the trivial topology of β-beryllene. The cubic trilayer phase also has a non-trivial Z_2 invariant. These results show that structural design and hydrogen functionalization provide direct routes for tuning the electronic, optical, and topological properties of two-dimensional beryllium.

V. DATA AVAILABILITY STATEMENT

The original data presented in this study are openly available in the GitHub repository `https://github.com/Photonico/H-Beryllene_20250718`.

VI. ACKNOWLEDGEMENTS

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